

Topic: 8086 Instruction Set

Q. Classify the Instruction Set of the 8086 microprocessor. Briefly explain Data Transfer, Arithmetic, Logical, Branch, String, and Processor Control instructions with one example each.

Definition to Remember

An **Instruction Set** is the complete collection of commands a microprocessor can understand and execute. The 8086 Instruction Set tells the CPU how to manipulate data in registers/memory, perform math, and control the flow of execution.

1. Classification of Instructions

A. Data Transfer Instructions

Moves data between registers or between memory and registers. (Does NOT affect flags).

* **Example:** `MOV AX, BX` (Copies the 16-bit content of BX into AX).

* **Others:** `PUSH`, `POP`, `IN`, `OUT`, `XCHG`.

B. Arithmetic Instructions

Performs mathematical operations. (Updates the Flag register).

* **Example:** `ADD AL, 05H` (Adds 05H to AL and stores the result in AL).

* **Others:** `SUB`, `MUL`, `DIV`, `INC` (Increment), `DEC` (Decrement).

C. Logical Instructions

Performs bitwise boolean operations and bit shifting.

* **Example:** `AND CX, DX` (Performs bitwise AND on CX and DX; stores result in CX).

* **Others:** `OR`, `NOT`, `XOR`, `SHL` (Shift Left), `ROR` (Rotate Right).

D. Branch / Control Transfer Instructions

Changes the normal sequential flow of execution by altering the Instruction Pointer (IP).

* **Unconditional Example:** `JMP 1200H` (Jumps directly to memory address 1200H).

* **Conditional Example:** `JZ LABEL` (Jump to LABEL if the Zero Flag is set).

* **Others:** `CALL`, `RET`, `LOOP`.

E. String Instructions

Optimized to manipulate blocks of data (strings/arrays) in memory.

* **Example:** `MOVSB` (Moves a string byte from Source Index (SI) to Destination Index (DI)).

* **Others:** `CMPSB` (Compare string), `SCASB` (Scan string).

F. Processor Control Instructions

Controls processor state or hardware synchronization.

* **Example:** `STC` (Sets the Carry Flag to 1).

* **Others:** `CLC` (Clear Carry), `CLI` (Disable interrupts), `HLT` (Halt).

★ Must-Write Points (for 10 marks)

1. An **Instruction Set** defines all commands the 8086 CPU can execute.
2. **Data Transfer**: Moves data (e.g., `MOV AX, BX`). Does not alter flags.
3. **Arithmetic**: Math operations (e.g., `ADD AL, 05H`). Updates flags.
4. **Logical**: Bitwise operations (e.g., `AND CX, DX`, `SHL`).
5. **Branch/Control**: Alters execution flow (e.g., `JMP`, `JZ`). Changes the IP register.
6. **String**: Operates on blocks of memory using SI and DI (e.g., `MOVSB`).
7. **Processor Control**: Manages hardware state/flags (e.g., `STC`, `HLT`).

⚡ Quick Recall

8086 Instructions → Data (MOV) → Arithmetic (ADD/INC) → Logical (AND/SHL) → Branch (JMP/JZ) → String (MOVSB) → Control (STC/HLT)